

Bryle Jaron Ong

858-705-0531 | ongbrylejaron@gmail.com | [linkedin.com/in/brylejaronong](https://www.linkedin.com/in/brylejaronong) | github.com/darksideshadow

EDUCATION

San Diego State University

San Diego, CA

B.S. Computer Science, B.A. Philosophy | Summa Cum Laude, GPA: 3.86

May 2026

EXPERIENCE

Autonomous Racing Capstone | *ROS, Linux, Gazebo, Python*

Aug. 2025 – May 2026

San Diego State University

San Diego, CA

- Integrated the full autonomous driving stack on a 1/10th scale racecar using ROS and Linux
- Configured a graph-based local planner for dynamic obstacle avoidance and overtaking maneuvers
- Presented progress and live demo to an audience at year-end as part of a 5-person team

Large Models for Code: A Review | *AI4BusinessLab*

Feb. 2024 – May 2024

San Diego State University

San Diego, CA

- Co-authored a survey paper reviewing large language models applied to code-related tasks including code generation, summarization, repair, and completion
- Surveyed encoder, decoder, and encoder-decoder architectures including CodeBERT, GraphCodeBERT, CodeGPT, and CodeT5
- Explored open challenges in code LLMs including long-context understanding, model interpretability, and computational resource constraints

Taco Bell | *Service & Food Champion*

Jul. 2022 – Dec. 2023

Carmel Mountain

San Diego, CA

- Packaged food for customers accurately and quickly, following guidelines for kitchen equipment and food safety
- Assisted customers with order-taking in the drive-thru and dining lobby with smooth operation
- Organized, maintained, and cleaned the dining area with utmost diligence and efficiency

PROJECTS

CalculatedCollapse | *Unity, C#*

May 2025

- Co-developed a multiplayer game in Unity with a 4-person team, contributing to player-side systems
- Implemented player movement, character models, and animations using C# scripting in Unity
- Developed input listeners for sound effect triggers and integrated player scripts with teammates' networking and UI systems

Two-Pass Assembler | *C++*

Nov. 2025

- Built a two-pass assembler in C++ for a systems programming course, translating SIC/XE assembly language into object code
- Implemented a line parser, opcode lookup table, and symbol table to resolve labels and instructions across two passes
- Generated object code and listing files as output, handling error detection throughout the assembly process

MercSquad | *Solidity, CSS, JavaScript*

May 2024

- Utilized HardHat, an Ethereum development environment, to create a two-player cryptocurrency card game
- Designed user GUI: added AI-generated fighter images, customized player icons, background, and audio
- Connected frontend user calls to backend smart contract deployed on Fuji test network

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, C, HTML/CSS, Go, MIPS, MySQL, Haskell, Prolog, Solidity

Frameworks: React, Node.js, Unity

Developer Tools: Git, Vim, VS Code, Eclipse

Libraries: STL, pthreads, NumPy, Matplotlib